

RUOXI LI

Department of Mathematics
University of California, Berkeley, California, United States
+1 6059403262 | email: ruoxi_li@berkeley.edu

EDUCATION

University of California, Berkeley
Doctoral Program in Mathematics
Research Interests: Algebraic Geometry

California, United States
AUG 2022 – present

Columbia University
BS in Applied Mathematics

New York, United States
August 2020 to May 2022

Augustana University
BA in Mathematics

South Dakota, United States
August 2017 to May 2020

PUBLICATIONS AND PREPRINTS

- *GIT for Root Stacks and 3D Mirror Symmetry*, with S. Garg, Y. Okitani, preprint;
- <https://arxiv.org/abs/2608.30350>
- *Minimal Trivializing Isogenies of Gm-Gerbes over Abelian Varieties and Period-Index Problem*, submitted;
- <https://arxiv.org/abs/2605.30530>
- *Point Objects and Derived Equivalences of Twisted Derived Categories of Abelian Varieties*, submitted;
- <https://arxiv.org/abs/2511.19282>
- *Weyl-invariant subspaces are not (usually) generic*, With R. Kurama, H. Talbott, R. Webb, submitted.
- <https://arxiv.org/abs/2510.03963>

TALKS/PRESENTATIONS

- *Point Objects and Twisted Fourier-Mukai Partners of Abelian Varieties*, Algebra Seminar, University of Oregon, 05/2026;
- *Ranks of Twisted Semi-homogeneous Vector Bundles and their Moduli Stacks*, Student Arithmetic Geometry Seminar, UC Berkeley, 03/2026;
- *Point Objects on Projective Bundles over Abelian Varieties*, Student Arithmetic Geometry Seminar, UC Berkeley, 11/2025;
- *Point Objects and Fourier-Mukai Partners of Twisted Derived Categories of Abelian Varieties*, University of Oregon, flash talk at WAGS 2025, 11/2025
- *Point Objects in the Twisted Derived Categories of Abelian Varieties*, Student Arithmetic Geometry Seminar, UC Berkeley, 04/2025;
- *Ample Vector Bundles on Tame Stacks*, Student Arithmetic Geometry Seminar, UC Berkeley, 10/2024;
- *The Fourier-Mukai Equivalences between a K3 Surface and its Moduli Space of Semistable Sheaves*, Student Arithmetic Geometry Seminar, UC Berkeley, 03/2024;
- *$D^b(X)$ determines X when K_X is ample or anti-ample*, Student Arithmetic Geometry Seminar, UC Berkeley, 04/2023;
- Final Presentation on GIT project at Michigan Research Experience for Graduates Symposium in July 2022.
- Poster presentation on Thick Points in a Random Square Subdivision Model Generated by Bernoulli Trials at Columbia University's Undergraduate Research Symposium in October 2021;
- Seminar Presentation on Foundations of Probability Theory at Columbia Undergraduate Seminar in Applied Mathematics in October 2021;
- Final presentation on Thick Points in a Random Square Subdivision Model Generated by Bernoulli Trials at Columbia University in August 2021.

WORK EXPERIENCE

University of California, Berkeley
Graduate Student Instructor

California, United States
January 2023 to present

- Math 16A, Analytic Geometry and Calculus, Fall 2026;
- Math 51, Calculus I, Summer 2026;
- Math 55, Discrete Mathematics, Fall 2025;
- Math 33, Precalculus, Fall 2024;
- Math 53, Multivariable Calculus, Summer 2024;
- Math 1A (51), Calculus, Fall 2023;
- Math 16B, Analytic Geometry and Calculus, Spring 2023.

Columbia University

Teaching Assistant

- Math GU4041, Introduction to Analysis, Spring 2022.

New York, United States

January 2022 to May 2022

Augustana University

Tutoring

South Dakota, United States

August 2018 to May 2020